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RESEARCH

Reservoir computing as digital twins for controlling nonlinear dynamical systems

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Abstract
Controlling complex and unknown chaotic systems remains challenging both theoretically and practically, especially when the intrinsic mechanisms of the system are elusive and/or the physical validation of the control strategies is infeasible. Developing effective and efficient control paradigms for these systems thus becomes an essential issue. In this work, we propose an approach using reservoir computing-based digital twins to replicate such unknown systems, providing a validation platform for the design of appropriate controls with only observables. Numerical experiments demonstrate the effectiveness of the proposed digital twins in controlling chaotic systems ranging from low- to high-dimensional cases, applying traditional control strategies including linear feedback methods and advanced nonlinear and time-delay techniques. We further investigate how to select and implement appropriate control strategies as well as their intensities according to specific requirements such as driving the system towards desired dynamics. This approach brings new insights into the field of data-driven control and offers new tools for designing practical control strategies.

Keywords Reservoir computing · Digital twins · Chaos control · Nonlinear dynamics

1 Introduction

Controlling chaos in dynamical systems is a focus of research that spans various disciplines, aiming to address the inherent unpredictable and seemingly random behavior of these systems [1–3]. The primary objective in chaos control is to stabilize or guide chaotic systems by applying suitable control signals [4, 5]. Chaos control research has found applications across various fields, including autonomous automobiles [6], biomedical signal processing [7], financial market forecasting [8], and environmental system prediction [9]. One critical challenge in controlling the chaotic behavior of systems in the physical world lies in their highly nonlinear nature, where slight variations of initial values and perturbations may lead to completely different chaotic dynamics [10, 11]. This typically leads to the following two distinct directions in chaos control research: suppressing detrimental chaotic behaviors and steering the systems from unstable chaotic states towards a predetermined fixed point or steady states [12], or harnessing beneficial chaos by directing the system towards specific periodic orbits or ordered chaotic states when chaotic behavior is useful in certain applications [13].

Many chaos control strategies have been proposed over the years, which can generally be categorized into open-loop and closed-loop control [14, 15]. Open-loop control refers to methods in which control signals or inputs are designed in advance, without relying on the current state or real-time output feedback of the system [16]. In contrast, closed-loop control leverages real-time feedback mechanisms to monitor the states or outputs of the system, and dynamically adjusts control signals according to the current feedback signal [17, 18]. Closed-loop control provides greater flexibility and can include various approaches such as linear feedback control [19], pyragas control [20], and

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global finite time control [21]. Although these strategies can achieve effective controlling results, they encounter challenges when applied to systems with higher dimensions [22, 23]. Most existing studies have not fully addressed the universality and adaptability of control strategies across different system dimensions and complexities [24]. For example, high-dimensional chaotic systems, such as the Lorenz-96 model, pose higher demands on control strategies due to their highly complex dynamic behaviors [25].

Effective implementation of these control strategies often necessitates continuous adjustment of various parameters [26, 27]. However, directly validating parameter adjustments on real systems is often impractical due to substantial costs and risks. In safety-critical domains—particularly when the controlled plant is a living organism—experimental interventions may entail surgical or physiological hazards [28, 29]. In such circumstances, a *digital twin* therefore provides a lower-risk validation platform to probe control designs before any physical intervention. In this work, a *digital twin* is a calibrated, data-linked computational representation of a specific physical system that reproduces its state evolution under candidate inputs [30]. Since accurate models of these systems are usually unavailable, a purely data-driven approach to construct digital twins demonstrates considerable advantages, particularly in cases lacking precise system knowledge [28, 31]. Consequently, despite potentially limited knowledge of the system's internal mechanisms, we can evaluate and optimize control strategies in a controlled and safe environment using the twin model [32, 33]. Once the control is achieved in the digital twin, the control signals can then be transferred to the nonlinear chaotic system to achieve the desired effects in the physical world. Thus, chaos control can be effectively implemented in a model-free and widely applicable approach.

A commonly adopted choice of the machine learning frameworks to build digital twins is the reservoir computing (RC), a variant of the recurrent neural network (RNN), which has enjoyed recent attention since the seminal works by Jaeger [34] and Maass [35], due to its excellent ability for model-free prediction of nonlinear and chaotic systems [36–39]. Its architecture comprises an input layer, a linear output layer, and a reservoir network consisting of sparsely connected dynamical neurons. While the input matrix and the reservoir's weighted adjacency matrix are randomly generated and fixed, training the output matrix occupies the entire computational expense, which can be efficiently obtained by least squares optimization [40].

Alongside purely data-driven approaches, there is also a rapidly growing strand of structure-preserving / physics-constrained machine learning which embeds physical priors directly into the model. Hamiltonian neural networks and symplectic networks conserve invariants (energy or symplectic structure) to improve long-horizon stability in Hamiltonian dynamics [41, 42]. In parallel, physics-informed neural networks (PINNs) that constrain training by governing equations, boundary/initial conditions and causality have shown remarkable results on chaotic benchmarks such as Lorenz and Kuramoto–Sivashinsky systems [43]. In a related hybrid direction, physics-constrained reservoir computing injects conservation or statistical constraints into RC and has achieved accurate short-term transients and credible long-term statistics in chaotic or turbulent flows [44]. These paradigms are compelling when system equations or structural constraints are available and physically consistent long-period behavior is utmost important, but typically demanding careful architectural design and heavier training. In contrast, our focus is on a model-free RC framework that remains effective when only time-series data are available, which enables lightweight, equation-agnostic digital twins for autonomous prediction and control experiments.

In this study, we specifically focus on constructing and controlling a digital twin solely from observational data gathered from a system with unknown internal dynamics. The core of this approach lies in utilizing the dynamic response characteristics of RC running in autonomous mode to simulate the behavior of the unknown system, thereby achieving accurate prediction and control of the system. By leveraging the high-dimensional dynamic mapping capabilities of RC, our approach is capable of embedding low and high-dimensional chaotic systems and achieving effective control. In addition, our method demonstrates robustness in the face of external disturbances. Various experiments have shown that the RC-based digital twin platform can be used to select appropriate control intensities and strategies as needed, enabling customized control over different chaotic systems.

2 Methods

This section outlines an overview of the methods employed in this paper, proceeding through the following steps. We first operate the original nonlinear system to gather data on its uncontrolled dynamic behaviors, which then serve as the training input for the reservoir computing model. After completing the input setup and other initial configurations, the standard RC training phase is conducted, allowing the model to determine the output matrix, which is the only indeterminate hyperparameter. The trained RC can be deployed in autonomous time-series prediction: given an initial input state $\boldsymbol{x}_0 \in \mathbb{R}^n$, RC iteratively generates state predictions, which forms the digital twin's baseline dynamics. In this mode, different control strategies can be applied by simply combining control signals with the predicted outcomes, which enables the evaluation and comparison of the control strategies and their strengths. Typically, the acquisition of training data entails uniform sampling, as equidistant

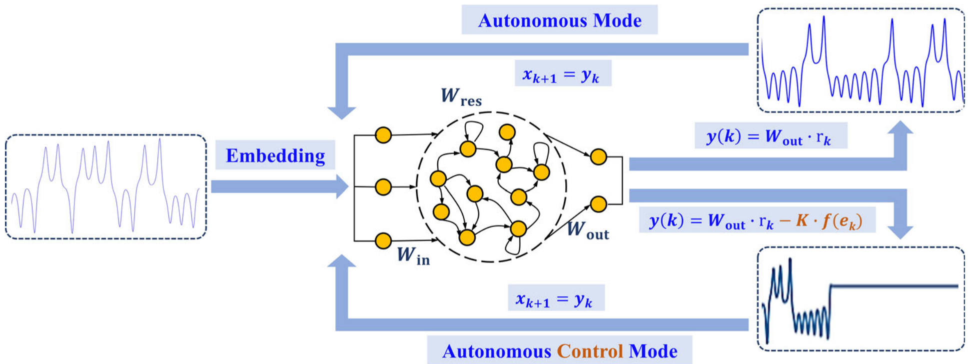

Fig. 1 **Autonomous mode and autonomous control mode of RC.** In autonomous mode, the reservoir computing model is configured for time series prediction, accurately forecasting the system's true dynamics, with blue and red lines denoting the true values and the predicted results respectively. In autonomous control mode, the RC implements the control of the time series by inputting the error between the target value and the output value. Here the control target is set to a point. This mode enables safe and convenient evaluation of control strategies and control intensity thresholds within the reservoir computing-based digital twin framework

time-series data are crucial to ensure that the standard reservoir computing model operating in autonomous mode can accurately predict the original system's future, which is a fundamental prerequisite underlying the present work. The approach is schematically illustrated in Fig. 1 .

2.1 Standard reservoir computing

The standard reservoir computing framework can be described in the state updating rule of the reservoir neurons [34]:

$$\begin{aligned} \boldsymbol{r}_k &= (1-\alpha)\boldsymbol{r}_{k-1} + \\ &\quad \alpha\phi(W_{\text{res}}\boldsymbol{r}_{k-1} + W_{\text{in}}\boldsymbol{x}_k + \boldsymbol{k}_c), \end{aligned} \tag{1}$$

where $\boldsymbol{r}_k \in \mathbb{R}^m$ represents the state of m reservoir neurons at time step k, and $\boldsymbol{x}_k \in \mathbb{R}^n$ is the input signal observed from a dynamical system φ evolving on a compact manifold $\mathcal{M}$. All state updates are executed at the same sampling step Δt, using uniformly spaced data. $W_{\text{in}} \in \mathbb{R}^{m\times n}$ and $W_{\text{res}} \in \mathbb{R}^{m\times m}$ denote the input weight matrix and the reservoir network matrix respectively, which are randomly generated and fixed according to certain distribution laws. $\alpha \in (0, 1]$ is the leakage factor and function ϕ determines the neuronal dynamics which at its simplest can be set as $\tanh(\cdot)$. $\boldsymbol{k}_c \in \mathbb{R}^n$ is the constant bias and is usually set to the all-ones vector. Consequently, the output $\boldsymbol{y}_k \in \mathbb{R}^l$ linearly combines the reservoir states such that $\boldsymbol{y}_k = W_{\text{out}}\boldsymbol{r}_k$, with only the output weight matrix $W_{\text{out}} \in \mathbb{R}^{l\times m}$ requiring training. RC can be adapted to different tasks. In the case of multi-step time series prediction [45–49], the objective is to make $\boldsymbol{y}_k = \boldsymbol{x}_{k+1}$ or to approximate it as closely as possible, thus W_{out} can be calculated by minimizing the loss function

$$\mathcal{L} = \sum_{k=1}^{N} s(k)||\boldsymbol{x}_{k+1} - W_{\text{out}}\boldsymbol{r}_k||^2 + \beta||W_{\text{out}}||^2, \tag{2}$$

where $s(1), s(2), ..., s(N)$ are weights and $\beta > 0$ is the L_2-regularization coefficient. Typically, the weights are assigned according to the standard "discard the transient" scheme, which is essentially $s(k) = 0$ for $k \leq W$ and $s(k) = 1$ for $k > W$. $W \geq 0$ is a washout length that omits the initial transient of the reservoir caused by the initialization r_0, during which the reservoir is driven by the input but those steps are excluded from the regression [50]. After the training phase, the output $\boldsymbol{y}_k$ can be redirected to the input layer, i.e., $\boldsymbol{x}_{k+1} = \boldsymbol{y}_k$, so that RC can run in autonomous mode (in this case $l = n$). This separation (teacher forcing for training while closed-loop for inference) follows standard RC practice and prevents exposure bias during training. With these settings, RC has been shown to accurately capture the dynamics of the input dynamical system. Meanwhile, the effectiveness of a single RC relies on a key microscopic property of the reservoir, namely the Echo State Property (ESP) [51]. Specifically, for any two different initial states $\boldsymbol{r}_0$ and $\boldsymbol{r}_0^{'}$, the states of reservoir neurons evolving under updating rule Eq.(1) asymptotically converge to an identical trajectory, i.e.,

$$\lim_{k\to\infty} ||\boldsymbol{r}_k - \boldsymbol{r}_k^{'}|| = 0. \tag{3}$$

A sufficient condition ensuring the ESP is that the spectral radius of W_{res} is smaller than 1. In practice, this requirement is typically achieved by globally rescaling the randomly initialized reservoir matrix W_{res}, such that its spectral radius is set to a value $\rho < 1$.

The Mean Squared Error (MSE) is used to evaluate the predictive performance of RC at different stages:

$$\mathrm{MSE}_k = \frac{1}{\tau} \sum_{p=k-\tau+1}^{k} ||\boldsymbol{y}_p - \hat{\boldsymbol{y}}_p||, \tag{4}$$

where τ denotes a specified time window, $\boldsymbol{y}_p$ and $\hat{\boldsymbol{y}}_p$ denotes the RC-predicted output and the corresponding state of the target system. A well-trained RC can efficiently achieve accurate prediction with saving computations. However, it is still important to note the standard RC model used here is finite-dimensional and does not explicitly encode or reproduce the infinite-dimensional state history of systems such as time-delayed systems; and handling time delays requires dedicated extensions (see Discussion and outlook).

2.2 Control methods and strategies

For the purpose of describing the control methods, we assume that the unknown system is formulated as Eq.(5):

$$\begin{cases} \dot{\boldsymbol{x}}_t = f(\boldsymbol{x}_t, \boldsymbol{u}_t), \\ \boldsymbol{y}_t = g(\boldsymbol{x}_t), \end{cases} \tag{5}$$

where $\boldsymbol{x}$ is the internal state of the system, $\boldsymbol{y}$ is the observable output of the system, and $\boldsymbol{u}$ is the accessible input. Generally, internal dynamics f and observation function g are unknown, and the only available information is the simultaneous response $\boldsymbol{y}_{\mathrm{train}}$ of the system to a certain input signal $\boldsymbol{u}_{\mathrm{train}}$ during a training period.

A control method typically involves determining a suitable control signal from the measured output $\{\boldsymbol{y}_t\}$ and a desired target $\{\boldsymbol{d}_t\}$. Broadly, one distinguishes *open-loop* schemes, which pre-specify an input sequence (often by solving an optimization problem over a finite horizon), from *closed-loop* (feedback) schemes, which compute the input from the current system output or tracking error. Closed-loop (or feedback) control methods are most commonly adopted in practice, which often provide more stable and robust results. Specifically, the feedback control signals we adopted are usually defined using the error $\boldsymbol{e}_t$ between $\boldsymbol{y}_t$ and $\boldsymbol{d}_t$, denoted by $f(\boldsymbol{e}_t)$, and is applied to the input in the next time step, given by $\boldsymbol{u}^*_{t+1} = \boldsymbol{u}_{t+1} + f(\boldsymbol{e}_t)$. Here, $\boldsymbol{u}^*_{t+1}$ denotes the updated input signal that is about to be fed into the system in place of $\boldsymbol{u}_{t+1}$. The ultimate goal of the control strategy is to ensure that the output of the next step $\boldsymbol{y}_{t+1}$ converges to the desired target $\boldsymbol{d}_{t+1}$, that is, $\boldsymbol{y}_{t+1} \to \boldsymbol{d}_{t+1}$.

Similarly, when employing the RC as a digital twin of an unknown system, the goal is to determine a new input signal $\boldsymbol{x}^*_k$ such that the RC exhibits the desired dynamical behavior, as illustrated in Fig. 1. In autonomous control mode, the control signal $f(\boldsymbol{e}_k)$ also needs to be fed back into the input layer just as in the original controlled system, i.e., $\boldsymbol{x}^*_{k+1} = \boldsymbol{y}_k - K \cdot f(\boldsymbol{e}_k)$, where K denotes the intensity of the control. Therefore, the reservoir state updating rule in the autonomous control mode is given by Eq.(6):

$$\begin{aligned} \boldsymbol{r}_k = (1-\alpha)\boldsymbol{r}_{k-1} + \alpha\phi[W_{\mathrm{res}}\boldsymbol{r}_{k-1} + \\ W_{\mathrm{in}}(\boldsymbol{y}_{k-1} - K \cdot f(\boldsymbol{e}_{k-1})) + \boldsymbol{k}_c]. \end{aligned} \tag{6}$$

Furthermore, a RC trained on observational data from a chaotic system is considered to be controlled if the target output $\hat{\boldsymbol{y}}_k$ and its observable output $\boldsymbol{y}_k$ in the autonomous phase satisfy $\lim_{k\to\infty} ||\boldsymbol{y}_k - \hat{\boldsymbol{y}}_k|| = 0$, where the superscripts denote the outputs generated by the RC and the target values. It should be noted that this serves as a performance criterion to assess control effectiveness, not as a constraint to guarantee convergence. In this case, the output of the RC is transformed from a chaotic trajectory into an unstable equilibrium point or unstable periodic orbits (limit cycles). It is important to note that control is a macroscopic concept that characterizes the dynamical relationship between the trained RC and the target system.

In this paper, we use three control strategies to generate different control signals: linear feedback control, offering a straightforward and intuitive method for adjusting the system's behavior; global finite-time control, which ensures rapid and accurate convergence of system states within a finite time; and Pyragas control, a time-delay feedback control strategy particularly effective at maintaining or stabilizing specific periodic orbits in chaotic systems. These strategies will be elaborated upon in the following sections.

After presenting the model, we address the key aspects of hyperparameter tuning of the RC model. There is a substantial body of RC literature on practical hyperparameter setting and we basically adhere to these practices. However, as hyperparameter design is not the main focus of this study, we refer interested readers to Appendix B for details about system-wise values and empirical working ranges. Instead, we summarize the hyperparameter tuning guidelines that underpin all subsequent results here. We view the leakage rate α as a timescale knob complementary to the sampling time step Δt of the original system. When the discretization already matches the signal timescale, the performance of RC is typically insensitive within a moderate range and α=1 is a often valid case. Consequently, we prioritize spectral radius ρ, input scaling, L_2 -regularization coefficient β and reservoir size m over fine-tuning α. In line with established RC practices, we perform a grid (or narrow manual) search over these hyperparameters, select settings that yield stable autonomous prediction and low short-horizon prediction error, and typically do not pursue an optimal value set.

3 Results

Before detailing results, we briefly outline the workflow by which the RC-based digital twin is used in this study. The RC model undergoes three phases: (1) training on the observational data to learn the original system dynamics; (2) prediction in autonomous mode (no control) to assess free-run fidelity; and (3) controller-in-the-loop evaluation in autonomous control mode, where the trained RC twin—with already fixed hyperparameters—is used to assess and compare control strategies. Accordingly, in the following subsection we now turn to a sequential discussion.

3.1 Verification of the RC-based digital twin

In the first stage of our numerical experiments, we first verify that RC can be employed to achieve a high-accuracy replication of an unknown chaotic system. The RC in autonomous mode is trained with the normalized x, y, z component of the benchmark Lorenz-63 system:

$$\begin{cases} \dot{x} = -\sigma x + \sigma y, \\ \dot{y} = rx - y - xz, \\ \dot{z} = -bz + xy, \end{cases} \tag{7}$$

with $\sigma = 10, r = 28, b = 8/3$, indicating that $O(0, 0, 0)$ is an unstable equilibrium point. In the experiments with the Lorenz-63 system, a training set comprising 10000 data points with a discrete time step of 0.025 is used, and the first 100 reservoir states are discarded to eliminate transient effects. Specifically as in Eq.(1), we set the reservoir size to $m = 1500$ neurons and the leakage factor $\alpha = 0.25$. The input matrix W_{in} and the reservoir matrix W_{res} are randomly generated and fixed with the elements in $[-1, 1]$. For the ESP mentioned in Ref. [51], W_{res} is rescaled to a spectral radius of 0.94. Unless stated otherwise, in all reported experiments we set a short washout length $W = 10$ because the training sequences are sufficiently long and the reservoir mixes quickly, resulting in insignificant net effect of a long washout for our tasks. Additionally, to ensure system stability and learning efficiency, appropriate L_2-regularization techniques are employed to prevent overfitting as in Eq. (2).

After training, the RC-based digital twin model operates in autonomous mode, accurately reproducing the dynamical behavior of the original Lorenz-63 system over an extended period, as illustrated in Fig. 2. However, in autonomous mode, prediction errors accumulate gradually, causing the RC dynamics to diverge from those of the original system after a certain time. This divergence moment, denoted by T_{cut}, serves as a critical benchmark to evaluate the prediction accuracy of the digital twin [52, 53]. Detailed information including definition of T_{cut} is provided in Appendix A.

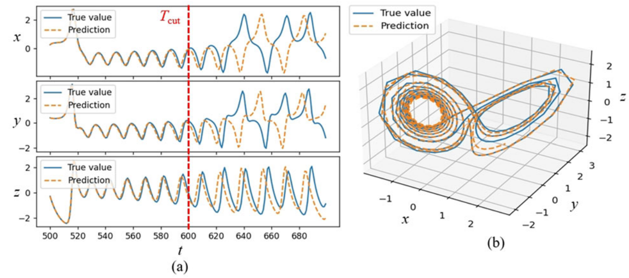

Fig. 2 Digital twin of the Lorenz-63 system. The predicted and true time series are shown in orange and blue curves, respectively. The time period from 1 to 500 is the data-driven training time, and the autonomous mode is entered at $t = 500$. The corresponding T_{cut} is approximately at $t = 600$

Similarly, for the h-dimensional Lorenz-96 system:

$$\frac{\mathrm{d}x_i}{\mathrm{d}t} = x_{i-1}(x_{i+1} - x_{i-2}) - x_i + f(t), \tag{8}$$

where $i = 1, 2, \cdots, h$ and $f(t)$ denotes a external periodic driving. Subsequent experiments confirm that for $h = 6$, its dynamics can also be precisely reproduced by the RC twin before T_{cut}. Due to the computational complexity involved in twining high-dimensional chaotic systems using a single large RC network, we adopt a parallel configuration [54], comprising several small-size RC networks, with each network responsible for a specific part of the target system. Overall, the accurate twinning before moment T_{cut} shows that it is feasible to use RC as a twin model for chaotic systems with unknown dynamics.

In the following sections, numerical experiments involving other chaotic systems, such as the Rössler and Chua systems, are conducted. Detailed information on parameter selections and specific experimental configurations in these experiments can be found in Appendix B.

Finally, to ascertain that the RC-based digital twin reproduces the defining chaotic properties of the target dynamics, we estimate the maximal Lyapunov exponents (MLEs, denoted by $\lambda_{\max}$) using the standard Wolf–Benettin framework across all four benchmark systems considered here (Lorenz–63, Rössler, Chua and Lorenz–96) [55, 56]. For the original continuous-time system , MLEs are computed along the full integration trajectory and reported per time unit, by simultaneously integrating the tangential variational equation and performing periodic QR reorthonormalization . For the discrete-time RC digital twin, we apply the same Wolf–Benettin procedure to the mapping's local linearization during the autonomous mode prediction (after training period) and likewise report values per time unit, avoiding artificially optimistic alignment and ensuring direct comparability with the original system. Writing $r_{t+1} = F(r_t)$ for the reservoir state update and $J_t = \partial F/\partial r|_{r_t}$ for the one-step Jacobian, we propagate orthonormal tangent directions

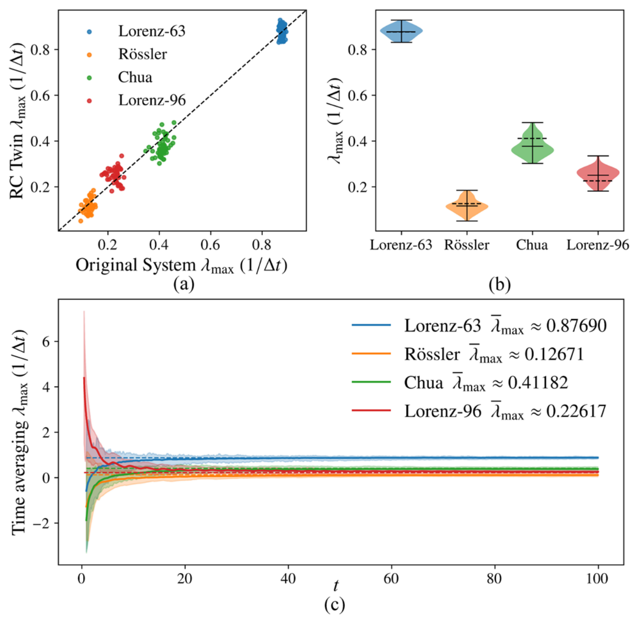

Fig. 3 **Maximal Lyapunov exponent (λ_{max}) validation of the RC-based digital twin across four chaotic systems.** (a) Paired consistency. RC twin MLE (vertical) versus original system MLE (horizontal) with points colored by system are presented, and the dashed line denotes equality (y=x). Each point corresponds to data acquired from an individual trajectory. (b) Distribution check. Violin plots of RC twin MLEs for each system. The dashed horizontal segments mark the mean original-system MLE (baseline). (c) MLE convergence in autonomous mode. Trail-wise mean time averaging RC twin MLE (solid) with a 95% confidence interval (shaded). The horizontal dashed lines still indicate the original baseline. All exponents are reported per time unit and window lengths are matched across systems

by $Z_t = J_t Q_t$ followed by $Z_t = Q_{t+1} R_t$ (QR reorthonormalization), and report the MLE per time unit as $\lambda_{max} \approx \frac{1}{n\Delta t}\sum_{k=1}^{n} \log R_{11}(t_k)$. In fact, the Jacobian of these cases can easily be derived as $J_t = (1-\alpha)I + \alpha\, D_t\, (W + k_{in} W_{in} W_{out})$ with $D_t = \mathrm{diag}\big(1 - \tanh^2(W_{res}\boldsymbol{r}_{k-1} + W_{in}\boldsymbol{x}_k + \boldsymbol{k}_c)\big)$. For each system, we perform multiple (over 40) independent experiments that differ in both training trajectory and random reservoir initializations, while the mean original-system MLE across experiments serves as a baseline reference against which the RC twin is evaluated, as shown in Fig. 3.

Across all four systems, the RC-based twin faithfully mirrors both the magnitude and the mean value of the MLEs observed in the original dynamics. In the paired consistency panel Fig. 3(a), the points concentrate basically around the identity line, indicating approximate one-to-one agreement at the trail level. Violin plots Fig. 3(b) show the twin's MLE distribution centered closely near the original baseline. And in Fig. 3(c), the cumulative time averaging MLE curves during the autonomous mode converge toward that baseline with narrow uncertainty bands, evidencing stable recovery of the MLEs. In every case the estimated maximal Lyapunov exponents are strictly positive for both the original dynamics and the RC-based digital twin, confirming chaotic behavior and indicating that the twin preserves this hallmark invariant. Moderate adjustments to numerical step sizes and autonomous mode window lengths within customary ranges do not alter these qualitative conclusions. We remind that, for starts in (or quickly attracted to) the same region covered by the training period, short-horizon prediction quality is essentially insensitive to the particular starting point and degrades at a rate according to the estimated MLE reported here, whereas long-horizon comparisons are made in terms of statistics rather than pointwise tracking.

After verifying finite-horizon reproduction and prediction, we use the RC-based digital twin as a validation and comparison platform for different control strategies; and the positioning relative to prior RC-twin studies is summarized in the Discussion.

3.2 Linear feedback control of RC-based digital twins

In what follows, we apply various control strategies to RC-based digital twins of different chaotic systems to validate the effectiveness of the proposed approach, beginning with linear feedback control. In a linear feedback controller, the error between the system's actual output and the desired output serves as the input signal fed back directly into the system, that is:

$$f(\boldsymbol{e}_k) = \boldsymbol{e}_k = \boldsymbol{y}_k - \boldsymbol{d}_k. \tag{9}$$

The point $O(0,0,0)$ is selected as the target state and the control intensity is set to $K = 1.5$ for the linear feedback control. We embark on a detailed examination of the system's response by applying control both before and after the critical time T_{cut}, enabling us to evaluate the effectiveness of the control strategy under varying system states.

In numerical experiments, we first investigate the RC-based digital twin of a Lorenz-63 system. As demonstrated in Fig. 4(a)(b), initiating linear feedback control at $t = 1000$ induces a rapid and distinct transition in the system's dynamics, leading to a convergence to the designated equilibrium point O. This holds true irrespective of whether RC operates within or beyond the range of precise prediction (before T_{cut}). Quantitative assessment of the resulting mean squared errors (MSEs) across the system's three dimensions confirms rapid stabilization, with MSE values settling around 0.007, 0.014 and 0.09 within only 15 time steps.

Further validation is provided by the digital twin of the more complex Lorenz-96 system, as depicted in Figure 4(c)(d). Upon activating the control at $t = 200$, a pronounced transition occurs, with the system swiftly steered toward the target equilibrium O. Despite exhibiting a more obvious perturbation initially and requiring a longer period

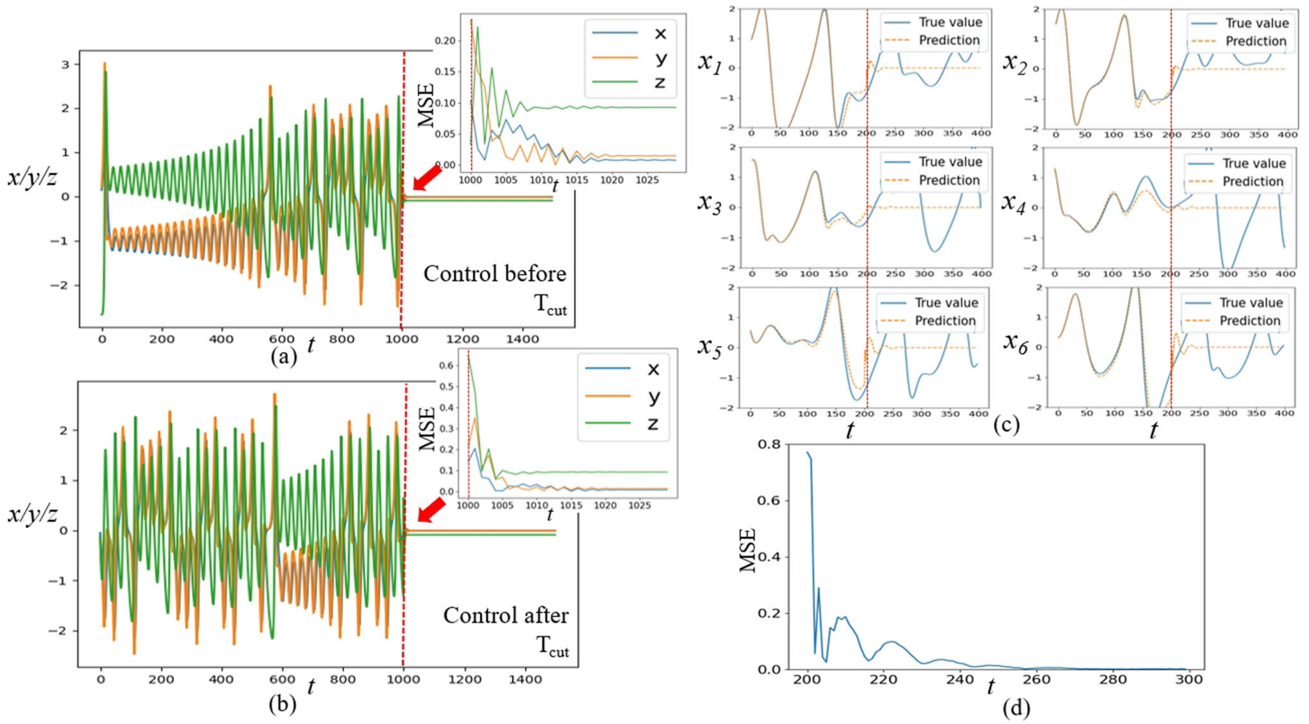

Fig. 4 Linear feedback control of the RC-based digital twin of Lorenz-63 and Lorenz-96 systems towards point O. Dynamics of the Lorenz system before and after the controller is switched on at (a)(b) $t = 1000$ for Lorenz-63 and (c)(d) $t = 200$ for Lorenz-96. The moment of applying control is (a) before T_{cut} and (b) after T_{cut}. For visual clarity, figure(c) separately shows the six-dimensional dynamics of the uncontrolled original system and the controlled digital twin. Inner panels and (d) panel are the corresponding MSEs after applying control

(approximately 60 time units) to reach equilibrium compared to the lower-dimensional system, the RC eventually stabilizes at an impressively low average MSE of approximately 0.003 across all six dimensions (see Fig. 4(d)).

To reiterate, we apply linear feedback control to the RC-based digital twins rather than the original chaotic systems themselves, and the results obtained from the digital twin closely approximate the known results of linear feedback control in stabilizing chaotic systems. These results illustrate the reliability when employing RC-based digital twins to evaluate linear feedback control strategies for chaotic systems.

3.3 Global finite time control of RC-based digital twins

Under identical test systems and configurations, we subsequently investigate the application of the global finite-time control method. In a global finite-time controller, when the system's state trajectory moves from outside to inside the unit sphere around the target, the control strategy switches from linear feedback control to finite-time control [21], that is,

$$f(\boldsymbol{e}_k) = I_{(||\boldsymbol{e}_k|| \leq 1)} \operatorname{sign}(\boldsymbol{e}_k) \cdot |\boldsymbol{e}_k|^s + I_{(||\boldsymbol{e}_k||>1)} \boldsymbol{e}_k, \tag{10}$$

where I_A denotes the indicator function of set A and $s \in (0, 1)$ is the steepness exponent. We still set the point O as the target and perform a comparative analysis relative to other control strategies.

For the low-dimensional Lorenz-63 system, we set the control intensity $K = 0.65$ and the steepness exponent $s = 0.8$. Still initiating the control at $t = 1000$, the RC system converges towards the target unstable equilibrium O within only 8 time steps, significantly faster than the linear feedback control scenario, as shown in Fig. 5(a)(b). With MSEs stabilizing at approximately 0.006, 0.013, and 0.092 in Fig. 5(a), respectively, global finite-time control validates faster convergence with the same accuracy, consistent with the results well known for general chaotic systems [21]. Notably, these experiments highlight a sequential control strategy, initially employing linear feedback control from $t = 1000$ to $t = 1003$, followed by finite-time control to optimally leverage the dynamic properties of the system and to steer the system more efficiently.

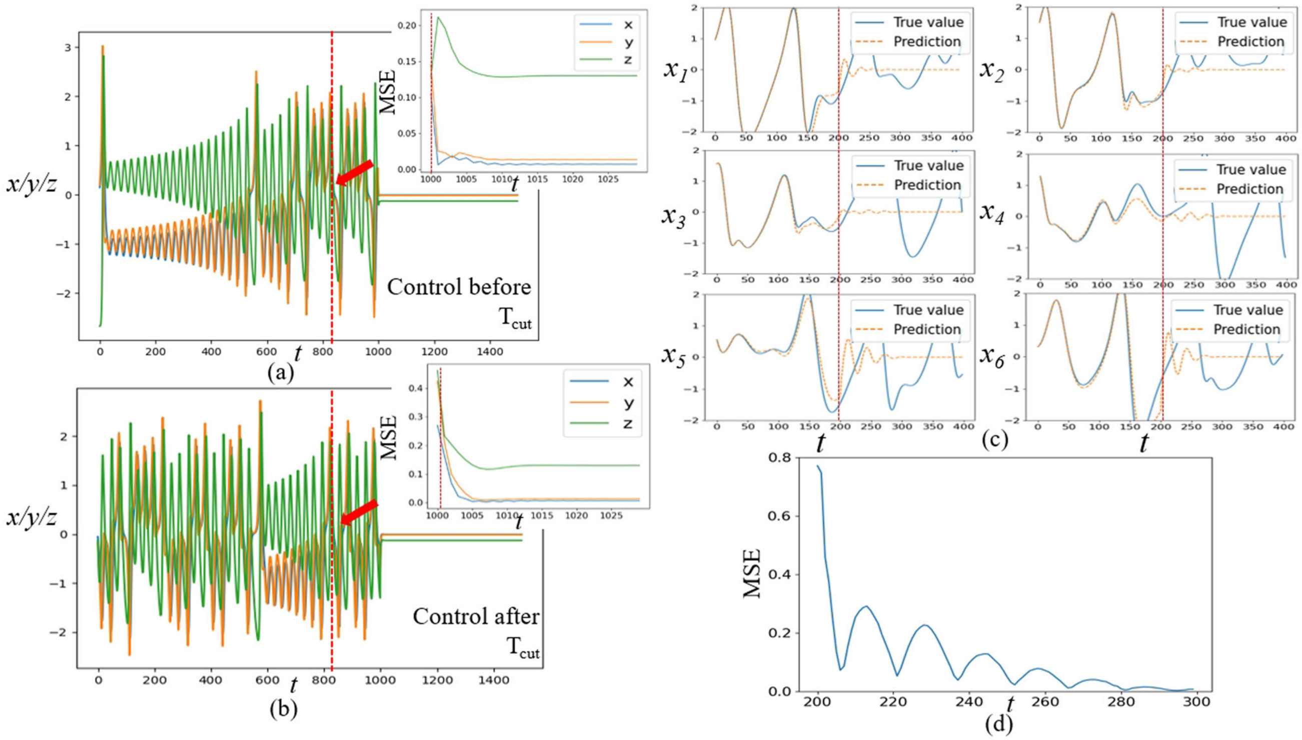

Fig. 5 **Global finite-time control of the RC-based digital twin of Lorenz-63 and Lorenz-96 systems.** Dynamics of the Lorenz system before and after the controller is switched on at (a)(b) $t = 1000$ for Lorenz-63 and (c)(d) $t = 200$ for Lorenz-96. The moment of applying control is (a) before T_{cut} and (b) after T_{cut}. Layout and presentation are identical to those of Fig. 4

Extending the analysis to the higher-dimensional Lorenz-96 system, we adjust the control intensity to $K = 0.3$, keeping the other settings identical to those used in the experiment shown in Fig.4(c)(d). As depicted in Fig. 5(c)(d), the RC-integrated system displays a gradual yet definitive convergence of its six dimensions toward the target point, marked by regular and notable smoothing in the magnitude of MSEs. Although this convergence requires approximately 100 time steps, it still leads to a precise stabilization, characterized by a wavelike decrement of the average MSE before stabilizing around 0.002 (see Fig. 5(d)).

It can be noticed that linear feedback control and global finite-time control achieved comparable stabilization outcomes using different control strengths. Therefore, we now carefully investigate the influence of the control intensity K on the control precision under these two distinct control strategies.

As illustrated in Fig. 6, there exists a clear correlation between K and the accuracy of convergence to the point O in the digital twins of the Lorenz-63 model. Generally, increasing the control strength consistently enhances convergence performance, as evidenced by the broadly progressive decrease of the average MSE across the three dimensions of the system. However, by comparing Fig. 6(a)(b), it is observed that, with linear feedback control, excessive control intensity K leads to higher average MSE and standard deviation across multiple experiments, whereas with global finite-time control, MSE remains consistently stable at a very low level throughout the experiments. These phenomena validates that global finite-time control exhibits higher stability

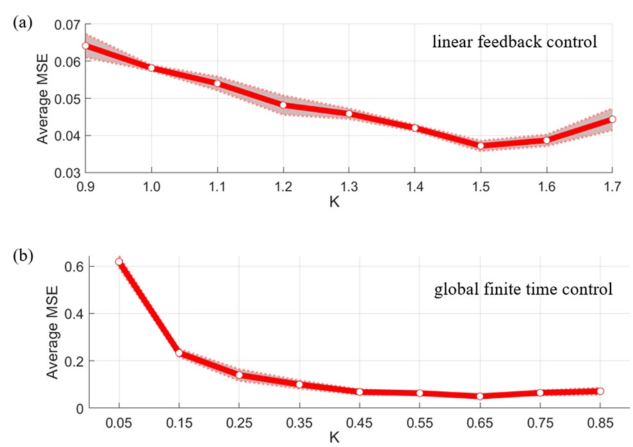

Fig. 6 **Effect of control intensity K on control effects of two control strategies on RC-based Lorenz-63**. As the control intensity K increases, the mean MSE of the three dimensions gradually decreases and is minimized at specific values. (a) corresponds to linear feedback control, (b) corresponds to global finite time control. The shaded area denotes standard deviations of 50 realizations

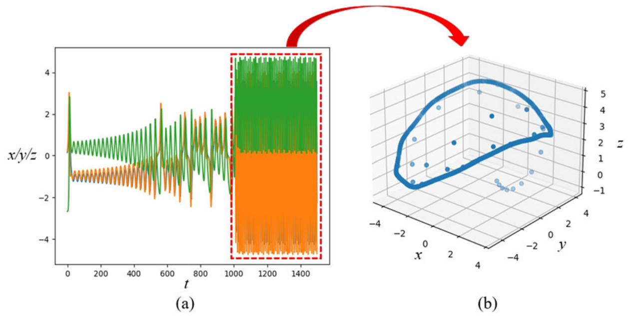

Fig. 7 Pyragas control of the RC-based digital twin of Lorenz-63 system to a periodic trajectory. (a) Dynamics of the Lorenz-63 system before and after the controller is switched on at $t = 1000$. (b) The corresponding trajectory after applying control for some time

and adaptability compared to linear feedback control, and can achieve effective control with lower control intensities across chaotic systems of varying dimensionalities. Based on the comparison with established results from the previous experiments, we confirm that the RC-based digital twin can function as a reliable platform to evaluate the performance and effectiveness of various control strategies applied to unknown chaotic systems.

3.4 Pyragas control of RC-based digital twins

Furthermore, we investigate the time-lag feedback Pyragas control strategy on the previously trained digital twins, revealing its distinct impact on system dynamics in comparison to linear feedback and global finite-time control strategies. The Pyragas control, noted for inducing periodic orbits in chaotic systems, offers a unique approach to controlling chaos. Unlike two control strategies previously discussed, a Pyragas controller does not require a specific target point; instead, its control signal is determined by the difference between the system's current state and its own delayed historical state, that is:

$$f(\boldsymbol{e}_k^*) = \boldsymbol{e}_k^* = \boldsymbol{y}_{k-1-\Delta k} - \boldsymbol{y}_{k-1}, \tag{11}$$

where Δk is the delayed time.

For the RC replicating the low-dimensional Lorenz-63 dynamics, we set the control strength $K = 0.65$ and delay $\Delta k = 1$. The control applied at $t = 1000$ results in rapid alignment of the system's trajectories into a periodic cycle, as clearly depicted in Fig. 7. Figure 7(b) demonstrates that, within 500 time steps post-control, the RC system's trajectories across the three dimensions consistently align along a cycle, illustrating the Pyragas control strategy's ability to transform chaotic motion into predictable periodic behavior.

Turning to the more complex Lorenz-96 model, with parameters set at $K = 0.6$ and $\Delta k = 1$, our empirical results reveal a more intricate system behavior under control. When

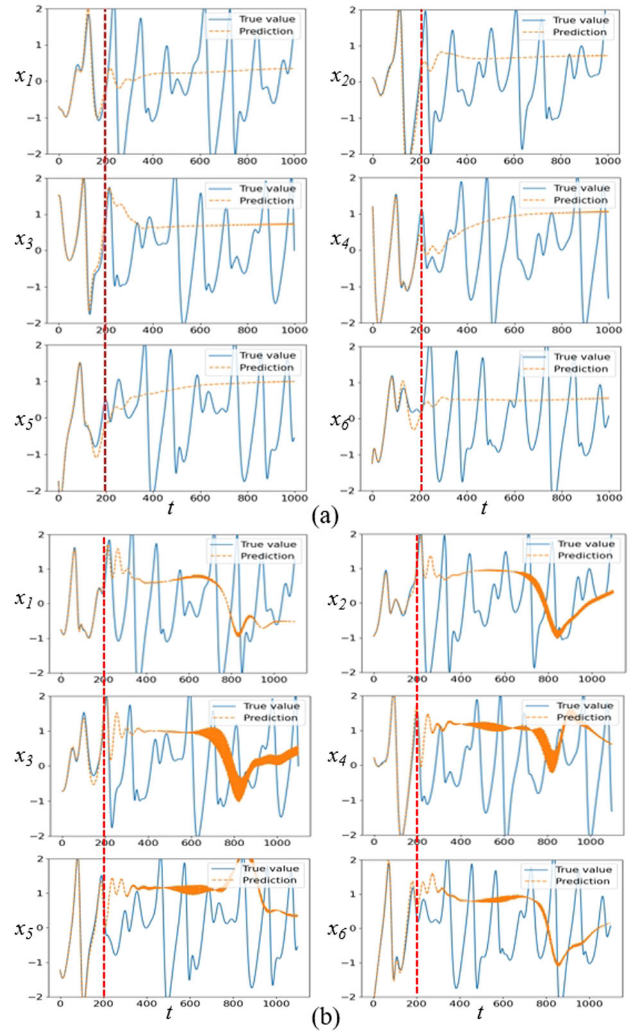

Fig. 8 Pyragas control of the RC-based digital twin of Lorenz-96 system. Dynamics of the Lorenz-96 system before and after the controller is switched on at $t = 200$ while control length set to (a) 800, (b) 900

the control is maintained for 800 time scales from $t = 200$, the system's dynamics are successfully pushed to a quasi-stationary state, with the final values across the six dimensions settling around 0.225, 0.668, 0.773, 0.764, 0.775, and 0.510, as depicted in Fig. 8(a). However, extending the duration of the control to 900 time steps significantly alters the dynamics. Under prolonged control, the efficacy of the Pyragas strategy diminishes, and the system's dynamics appears to return to a chaotic state. This observation highlights a critical aspect of the Pyragas control strategy: its effectiveness is closely related to the control duration, emphasizing the need for precise timing in its application.

The subtle interplay between control strength K and the resulting control outcomes triggers further detailed research and analysis of this correlation. Unlike the relationship observed in linear feedback and global finite-time controls,

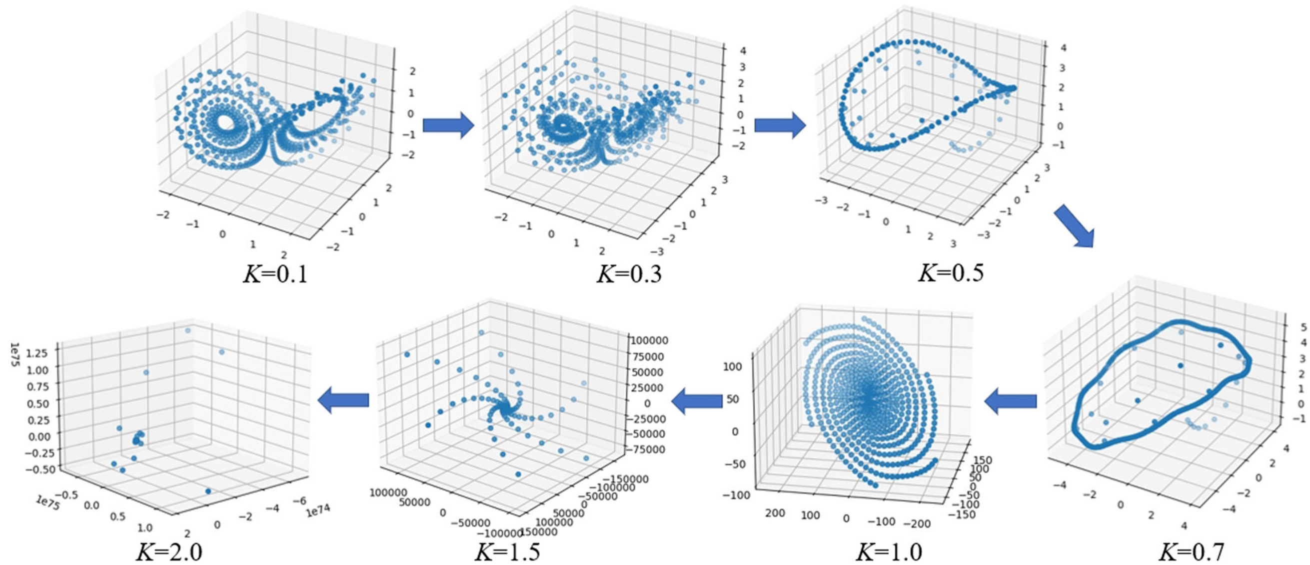

Fig. 9 **Effect of control intensity K on control effects of Pyragas control on RC-based Lorenz-63.** As the control strength K increases, the figure shows a representative attractor for the digital twin. As the control strength increases, there is a shift between the periodic orbit and the stationary point

the Pyragas strategy demonstrates a complex dependency on K, due to its more elaborate manipulation of the dynamics. As illustrated in Fig. 9, this relationship reveals a fascinating dynamical transition within the digital twin system.

With a moderate increase in K from 0, the system is initially directed into a periodic orbit, demonstrating the unique ability of Pyragas control to induce cyclic behavior in chaotic systems. This phase represents a controlled state where the system's dynamics is regularized into predictable periodic trajectories, a significant departure from chaotic behavior. As K further increases, the control effect undergoes a significant transformation. Beyond a critical threshold of control strength, the system transitions from the periodic orbit to asymptotic attraction to the equilibrium point O. This evolution emphasizes the inherent two-stage control capability of the Pyragas strategy, where varying K dictates the nature of the control outcome, whether periodic or stationary. It also highlights the method's versatility in targeting distinct dynamical states within the same system.

This intricate relationship between control strength and system behavior under Pyragas control illuminates the nuanced mechanisms through which control strategies can be optimized to achieve desired outcomes in complex systems. It also emphasizes the importance of precise parameter tuning in leveraging various distinct features of such control strategy, from inducing regular periodic orbits to achieving pinpoint convergence on stationary states.

3.5 Noise robustness of chaotic control strategies

In practical applications of control strategies, external disturbances are inevitable and often degrade the performance of the control system. To simulate the real-world scenario, we introduce Gaussian noise into the input of RC during the control process, aiming to examine its effect on control performance. Gaussian noise, a common type of random disturbance, effectively simulates the noise encountered in real-world environments, which makes this experimental setup reasonable for evaluating the robustness of control strategies in practical applications.

As shown in Fig. 10, for the RC embedded with the Lorenz-63, the control effect of the RC remained largely unaffected when Gaussian noise is added to the control inputs. In addition, we perform a similar noise robustness experiment on the Lorenz-96 system and obtain consistent conclusions. These observations show the robustness of our RC-based control approach applying the above control strategies, which maintains its effectiveness and stability in the presence of external disturbances. However, compared to the noise-free case, we still observe a slight degradation in system performance. This suggests that although the RC-based approach can effectively cope with Gaussian noise, algorithm optimization should still be considered to further improve the adaptability and robustness of the system in complex environments.

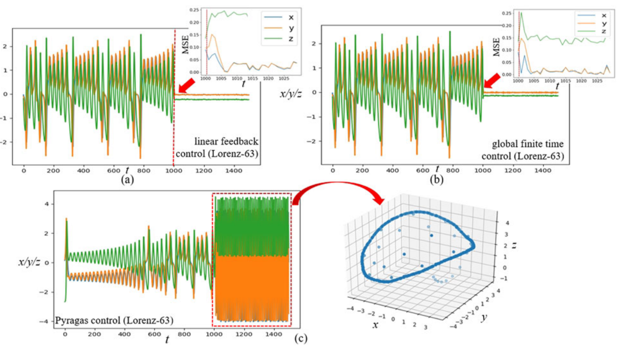

Fig. 10 Effect of noise on the control performance of the RC embedded with Lorenz-63 system. (a)(b) Panels show the imposition of linear feedback control and global finite-time control to RC with Gaussian noise introduced, and the inset in (a)(b) depict the MSEs of the system after the control is applied. (c) The left panel shows the imposition of the Pyragas control, and the right panel shows the periodic orbit after the control is applied

3.6 Effect of sampling step on control performance

The Methods section establishes that the RC model is trained and run on uniformly sampled, discrete-time data with step Δt. As this interface is standard in digital control, a practical issue is similarly highlighted: how does the choice of Δt affect closed-loop control performance in our autonomous control mode? To make this dependence explicit, we carry on the following experiments on the Lorenz-63 system.

We vary Δt while controlling what RC observes during training. Two simple protocols are used here to address the using of different time steps. In the fixed-time (FT) protocol, the physical training period T_{train} is fixed as Δt increases, thereby reducing the number of training samples but preserving the time span. While in the fixed-samples (FS) protocol, the number of training samples N_{train} is fixed, thereby extending the covered duration. For these two protocols, multiple parallel experiments are conducted for both linear feedback control and global finite-time control, with a fixed RC setting, the same control target $O(0, 0, 0)$ and control intensity. Figure 11 shows the mean curve of the settling time and RMS error computed on a late evaluation window versus Δt with a shaded 95% confidence interval under the FT and FS protocols.

As Δt decreases, RC's response to control signals has become more rapid and sensitive, which in turn shortens the post-control settling time and reduces the steady-state error for linear feedback control. The global finite time control exhibits a stronger sensitivity at very small Δt and benefits from control intensity choices that are consistent with the sampling step to avoid over-actuation, yet its overall trend with respect to Δt aligns with that of the linear feedback control. Across both protocols, we observe broadly consistent trends within the explored range: increasing Δt is

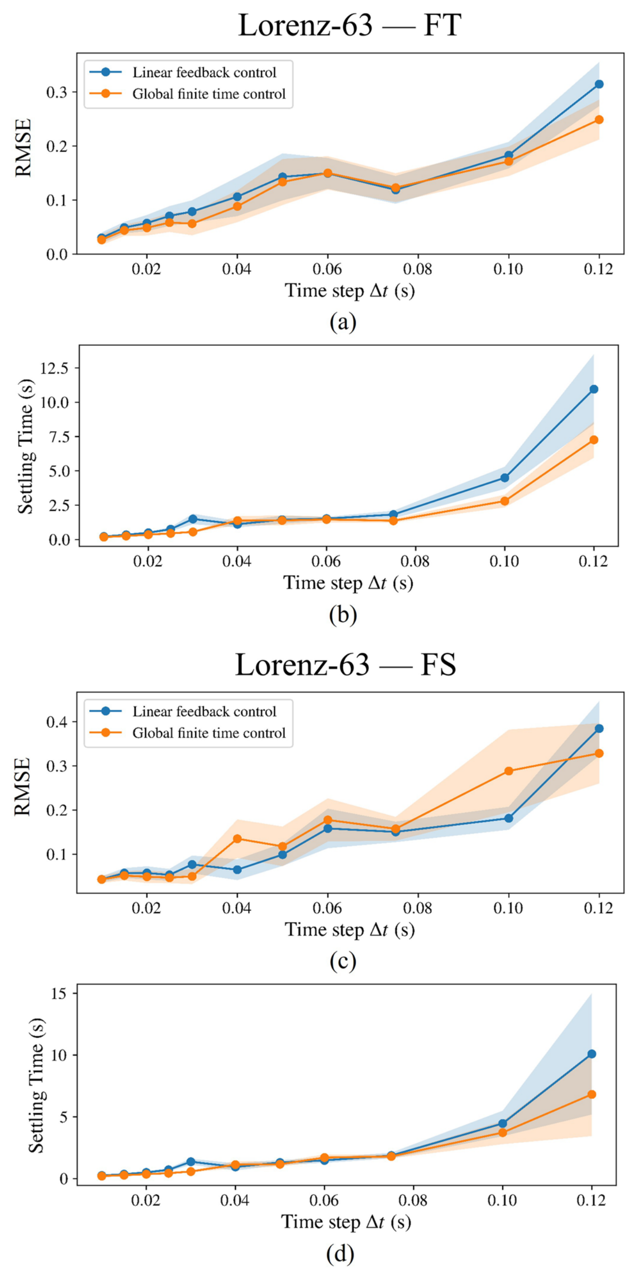

Fig. 11 Effect of sampling time step Δt on control performance of the RC embedded with Lorenz-63 system. Fixed window RMSE and settling time versus different Δt are shown for (a)(b) fixed-time protocol, and (c)(d) fixed-steps protocol. Shaded regions denote 95% confidence intervals of the mean. Across both protocols, a broadly similar trend is visible: larger Δt often coincides with degraded control performance. While the uncertainty bands indicate that these patterns should be interpreted qualitatively rather than strictly quantitatively

often associated with the expected degradation in closed-loop control behavior. Importantly, the uncertainty bands show non-negligible variability across different random initializations, so these patterns are intended as qualitative rather than quantitative conclusions. Overall, the plots offer an expected, protocol-consistent view of how discrete-time

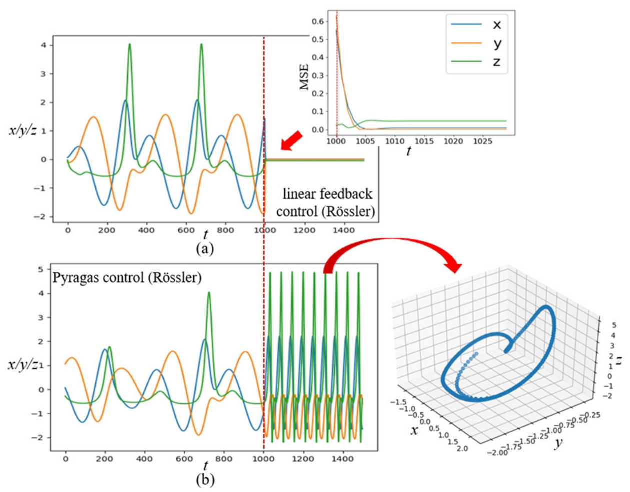

Fig. 12 **Linear feedback control and Pyragas control on the RC-based digital twin of a Rössler system.** (a) The system dynamics after applying linear feedback control to the RC embedded with Rössler. The inset illustrates the MSE of the system after the control is applied. (b) The left panel shows the system dynamics after applying Pyragas control, and the right panel shows the 3D orbit after applying the control

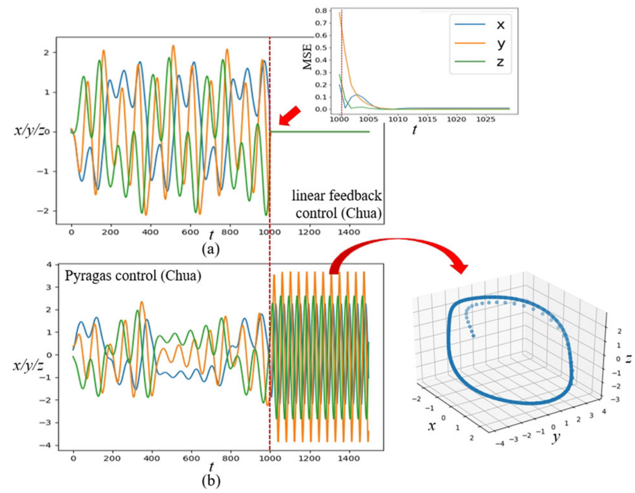

Fig. 13 **Linear feedback control and Pyragas control on the RC-based digital twin of a Chua system.** Layout and presentation are identical to those of Fig. 12; each panel and inset shows the corresponding results for the Chua system

sampling interacts with the control performance of the RC-based digital twin over the practical range studied here.

3.7 Experiments of other chaotic systems

To further validate the applicability and effectiveness of our RC-based control approach across different chaotic systems, we examine the Rössler system and the Chua system, two classical chaotic systems widely studied in chaos dynamics. The dynamics of the Rössler system are described by:

$$\begin{cases} \dot{x} = -y - z, \\ \dot{y} = x + ay, \\ \dot{z} = -b + z(x - c), \end{cases} \tag{12}$$

with $a = 0.5, b = 2, c = 4$, while the Chua system is expressed as:

$$\begin{cases} f = m_0 \cdot x + 0.5 \cdot (m_1 - m_0) \cdot (|x+1| - |x-1|), \\ \dot{x} = p \cdot (-x + y - f), \\ \dot{y} = x - y + z, \\ \dot{z} = -q \cdot y, \end{cases} \tag{13}$$

with $p = 10$, $q = 14.87$ and $m_0 = -0.68$, $m_1 = -1.27$. Details about experimental settings are provided in Appendix B.

In our experiments, we test the effectiveness of linear feedback control and Pyragas control on the RC-based Rössler system (see Fig. 12) and Chua system (see Fig. 13). It can be observed that the linear feedback control precisely stabilizes the system dynamics at the target point O, and the Pyragas control successfully steers the system dynamics to the desired periodic orbits. The above results further demonstrate the universality of the RC-based control approach.

4 Discussion and outlook

In this study, we address the intricate challenge of validating control strategies in real-world complex systems. By adopting a purely data-driven digital twin framework based on reservoir computing (RC), we propose an effective approach for modeling and testing various control strategies on chaotic systems in a safe and efficient manner. This approach can help practitioners circumvent the high costs and subject-level risks associated with real system experiments, offering a safer staging ground for controller tuning prior to deployment. Our method is validated through a series of experiments across different chaotic systems, both low-dimensional and high-dimensional, in order to evaluate its effectiveness and adaptability.

Our research begins by verifying RC's capability in accurately modeling system evolution thereby serving as a digital twin. Regarding sensitivity to initial conditions, the validity of the RC-based twin for chaotic systems primarily hinges on the training horizon covering the attractor: once the training trajectory adequately explores the invariant set, autonomous prediction remains reliable for different initial conditions whose transients enter and evolve on the same attractor. In practice, if operation targets states or regimes outside the attractor represented in the training data (e.g., distinct basins or qualitatively different dynamics), additional data collection and retraining are warranted. Next, the linear feedback and the global finite-time control strategies demonstrate robust performance in our framework when guiding the system towards a stationary point. The Pyragas control strategy is also evaluated for its efficiency in inducing desired periodic orbits in the RC-based digital twin. Meanwhile, we also investigate the influences of control intensity on performance—specifically, on aspects such as accuracy and periodic behaviors, and validate that global finite-time control outperforms linear feedback control through parallel experiments. Moreover, our results demonstrate that the Pyragas control can induce distinct periodic behaviors under different control intensities, consistent with known results. Furthermore, we assess the robustness and universality of our approach across diverse chaotic systems.

As our results focus on control-oriented use of RC-based digital twins, it is helpful to position this study relative to recent RC twin work by Kong *et al.* that emphasizes forecasting and monitoring [31]. In particular, the following remarks clarify how the present scope complements that literature while remaining consistent with our empirical findings. Building on the foundation of their work, the present study targets a complementary objective centered on control and contributes: (i) a controller-in-the-loop validation and comparison testbed, in which the RC twin is used to tune parameters and compare control strategies across several chaotic benchmarks, including noisy settings; the way in which control is applied on the RC twin is compatible with its application on physical systems, which increases confidence that conclusions drawn on the twin carry over to real systems; (ii) an explicit examination of operation after the twin's free-run prediction loses time-accuracy (T_{cut}), showing that the framework remains useful both before and after divergence; and (iii) a technical clarification: whereas prior work injects inputs(driving forces) using a random matrix during prediction, it does not develop or assess controller-in-the-loop procedures on the twin—here we articulate and validate such a workflow.

In conclusion, this study not only elucidates the underlying mechanisms of chaos control via the RC-based digital twin but also validates its reliability and convenience as a verification platform. Through multiple experiments, we demonstrate that the RC-based digital twin can be used to effectively compare the performance of various control strategies, facilitating the selection of appropriate controls. Additionally, this approach allows for testing and identifying suitable control intensities to achieve specific control purposes, such as steering chaotic systems toward desired periodic orbits via time-delay control. Our findings provide new and safe methods for achieving appropriate controls in chaotic systems, offering possible directions for future advancements in this field. The insights gained from this research contribute to the broader discussion on complex system control, providing interesting ideas for developing more sophisticated and adaptable control strategies.

Future works include enhancing the generalization capability and adding theoretical validation of the proposed framework. As previously noted, incorporating time-delay systems presents a challenging yet valuable extension of our current framework. Promising new architectures, such as infinite-dimensional Next-Generation Reservoir Computing (NG-RC) and neural delay differential equations, have recently been demonstrated to be effective tools for modeling time-delayed dynamical systems [57, 58]. A promising direction for future work may be to integrate such methods into our digital twin framework—replacing the standard reservoir computer used here—and to adopt a similar workflow to evaluate control strategies across various time-delay systems. Other commonly used control strategies, including open-loop optimal control, will also be considered.

Appendix A Details of the divergence moment T_{cut}

Here, we present the definition of T_{cut} omitted from the main text and provide a detailed exposition of it. T_{cut} is a forecasting diagnostic used to indicate when an autonomous (free-run) prediction of the RC-based digital twin departs from the ground truth; but it plays no operational role in controller evaluation in this study. Concretely, we formalize T_{cut} using the widely used "valid prediction time" criterion in the chaos/RC literature: the earliest time when a sliding-window normalized RMSE exceeds a preset threshold (with a short persistence to avoid jitter). This is consistent with recent practice in evaluating forecasts of chaotic systems [52, 53]. Namely, we define the sliding-window NRMSE as follows:

$$\begin{aligned} E(k) &= \frac{\sqrt{\frac{1}{w}\sum_{j=k-w+1}^{k}\|\tilde{y}_j - y_j\|_2^2}}{\sqrt{\frac{1}{w}\sum_{j=k-w+1}^{k}\|\tilde{y}_j - \bar{y}_{k,w}\|_2^2}}, \\ \bar{y}_{k,w} &= \frac{1}{w}\sum_{j=k-w+1}^{k}\tilde{y}_j, \end{aligned} \tag{A1}$$

where $\tilde{y}_k$ denotes the ground truth state of the original nonlinear system and y_j is still prediction of the digital twin at step k. Then T_{cut} is set as:

$$T_{\mathrm{cut}} = \min\{k : E(k) \geq \varepsilon \text{ for } \kappa \text{ consecutive steps}\}. \tag{A2}$$

Typical values for threshold ε is like $\varepsilon = 0.5$. In the main text, the figures labeled "before/after T_{cut}" are based on a manual inspection aligned with the same criterion.

Nevertheless, it should be pointed that T_{cut}, which is only used in experiments to mark whether control is switched on while the twin is still time-accurate or after it has diverged, is often unknown in our anticipated scenario and does not enter the controller design or tuning. Indeed, it is precisely to demonstrate that the framework presented in this work remains effective even without prior knowledge of T_{cut}, which is exactly what we desire to show.

Appendix B Details of the experiment settings

We provide a detailed description of the experiment configurations omitted in the main text with empirical working ranges, as well as the lightweight search routine we use in practice. The aim is reproducibility and practical guidance rather than the pursuit of a single "optimal" configuration. This begins with the parameters for the autonomous mode of RC embedded with the Lorenz-96, Rössler, and Chua systems. In Eq. (8), the driving force $f(t)$ is set to a sinusoid wave described by $f(t) = 2.2\sin(2t) + 2$ for inducing chaotic behavior. Hyperparameters of RC are selected by a small grid or narrow manual sweep over $(\rho,\ \alpha,\ \beta,\ m)$ under fixed preprocessing and input normalization. We evaluate candidate settings using (i) short–horizon prediction error on a held–out slice of the training period and (ii) stability of autonomous mode prediction over the same window. Consistent with established ESN/RC practice, we report a final setting together with an empirical working range.The latter indicating nearby configurations that performed comparably well on our setups, so that practitioners can trade off accuracy, robustness, and compute without exhaustive search [50]. Table 1 shows the parameters of different RCs to twin the four systems mentioned above, and the configurations remain the same in autonomous mode and autonomous control mode. Table 2 summarizes the ranges within which we observed comparable performance on our setups. The elements of matrix W_{res} are randomly generated according to a uniform distribution in $[-1, 1]$, and then rescaled to achieve the required spectral radius. The matrix W_{in} is similarly generated by uniform distributions but not deflated.

As mentioned in the main text, sampling the original system with an appropriate time step does have an impact on hyperparameter selection. Studies show that excessively coarse sampling degrades autonomous regeneration of chaotic dynamics, while overly dense sampling can also be suboptimal, indicating a window of suitable sampling frequencies [59]. In our experiments, we observed a similar pattern: when the time step grows beyond a system–dependent threshold (e.g., for Lorenz-63 system roughly 0.1∼0.15), reservoir training/prediction becomes really fragile across most hyperparameter choices. Within a reasonable range, shorter time steps generally broaden the robust region of selected hyperparameters and increase the success probability across random initializations, yet diminishing returns appear as the step becomes too fine—consistent with the non–monotonic behavior reported in [59].

As to the choice of the control intensity K, it is application–dependent and guided by the intended behavior rather than by a universal rule. In our examples with Pyragas control, varying K can lead to distinct periodic responses (e.g., stabilization of different orbits), and the practically useful region therefore depends on the target. In this paper we select K by manual tuning, and do not pay attention to optimal control intensities.

Table 1 Reservoir computing settings including reservoir size m, leakage factor α, L_2-regularization coefficient β and spectral radius ρ of reservoir matrix W_{res}

Model	m	α	β	ρ
Lorenz-63	1500	0.8	1e-08	0.94
Lorenz-96	$1200 \cdot 6$[1]	0.33	1e-12	0.75
Rössler	1000	1	1e-08	0.9
Chua	1500	1	1e-01	0.94

[1]Parallel configuration with 6 separate RC is used

Table 2 Empirical working ranges of reservoir computing settings on our setups

Model	m	α	β	ρ
Lorenz-63	[1000,1500]	[0.25,1]	1e-08	[0.94,0.99]
Lorenz-96	$1200 \cdot 6$	[0.3,0.4]	1e-12	[0.48,0.75]
Rössler	[1000,1200]	[0.8,1]	1e-08	0.9
Chua	[1200,1500]	[0.9,1]	1e-01	0.94

Author Contributions S.Y.L. conceived the idea; H.R.S., X.Z.Z., R.Z.C., and S.Y.L. designed the research; H.R.S. and X.Z.Z. performed the research; H.R.S., X.Z.Z., R.Z.C., J.W.H., C.G., and S.Y.L. analyzed the data and wrote the paper.

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Data Availability No datasets were generated or analysed during the current study.

Declarations

Conflicts of Interest/Competing Interests The authors have no relevant financial or non-financial interests to disclose.

Competing interests The authors declare no competing interests.

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